

3.10 Deep Learning on Tabular Data for Fraud Detection under Extreme Class Imbalance [PJT10]

Domain: Tabular data and financial technology. Suggested level: introductory to intermediate.

Background

Fraud detection combines two conditions that break naive deep learning practice: fewer than one positive case in every thousand transactions, and a data-generating process that shifts as fraudsters adapt. Gradient boosted trees remain a very strong baseline on tabular data, so any claim that a neural architecture is preferable needs careful support. This project tests that claim honestly and studies the cost-sensitive threshold decision.

Research questions

RQ10.1 Under what conditions, if any, do tabular neural architectures such as TabNet, FT-Transformer and SAINT outperform tuned gradient boosted trees on fraud detection?

RQ10.2 How do resampling, focal loss and cost-sensitive learning compare as strategies for extreme class imbalance, and how does each affect calibration?

RQ10.3 How quickly does model performance decay when evaluated on transactions from a later time period than the training window?

Datasets

- IEEE-CIS Fraud Detection on Kaggle, roughly 590,000 transactions with rich categorical and temporal features.
- Credit Card Fraud Detection on Kaggle (mlg-ulb/creditcardfraud), 284,807 transactions with 0.17 percent positives.
- The PaySim synthetic mobile money dataset on Kaggle, which is useful for mobile money scenarios relevant to African markets.

Suggested methods and baselines

Establish a tuned LightGBM or XGBoost baseline first. Then evaluate FT-Transformer, TabNet and a simple embedding-based multilayer perceptron. Use temporal splits rather than random splits so that leakage through time is impossible.

Evaluation plan

Area under the precision-recall curve, precision at a fixed alert budget, expected monetary cost under an explicit cost matrix, calibration assessment, and a performance-over-time plot showing decay.

Skills developed: Tabular modelling, imbalanced learning, cost-sensitive evaluation, temporal validation.

4. Common Methodological Expectations

The following requirements apply to every project in this document. They are stated once here so that each brief remains short, and students should treat them as part of the assessment criteria.

4.1 Baselines before architectures

Every project must include at least one simple, well-tuned baseline: a linear or tree-based model for tabular and time series work, a bag-of-n-grams classifier for text, and an ImageNet-pre-trained backbone with a linear head for vision. A deep model that cannot beat a tuned baseline is a finding worth reporting, not a failure to hide.

4.2 Splitting and leakage

Splits must respect the structure of the data. Split by patient in clinical work, by speaker in speech, by geographic block in remote sensing, and by time in forecasting and fraud detection. Random splits at the level of individual records inflate performance in almost every project listed here.

4.3 Calibration and uncertainty

Where a model output will inform a decision, students should report calibration alongside discrimination. Reliability diagrams and expected calibration error take very little additional work and often change the interpretation of a result substantially.

4.4 Subgroup and worst-case reporting

Aggregate metrics conceal uneven performance. Students should report results disaggregated by language, site, region, demographic group or acquisition device as appropriate to the project, and should state the worst-group figure explicitly rather than only the average.

4.5 Compute, cost and reproducibility

Report training time, hardware used and approximate energy or monetary cost. Fix and report random seeds, run at least three seeds for headline comparisons, and report variability rather than a single number. All code should be released in a repository with an environment specification and a script that reproduces the main table.

4.6 Ethics, licensing and data governance

Students must check the licence of every dataset, obtain ethical clearance where human subjects data is involved, and state clearly what the model should not be used for. Clinical datasets require a data use agreement, and social media datasets carry re-identification risk even after anonymisation. A short data statement describing provenance, annotator demographics and known limitations should accompany every submission.

5. Suggested Project Timeline

Period	Focus
Weeks 1 to 3	Literature review, dataset acquisition, licence and ethics checks, reproduction of one published baseline. [28/08/2026]
Weeks 4 to 6	Data pipeline, exploratory analysis, simple baseline model, agreed evaluation protocol and split strategy. [11/09/2026]
Weeks 7 to 11	Main experiments, ablations, hyperparameter search with a documented budget.[09/10/2026]
Weeks 12 to 13	Robustness, calibration and subgroup analysis, error analysis, negative results written up. [16/10/2026]
Weeks 14 to 15	Report writing, code release, reproducibility check by a peer. [23/10/2026]

6. Note on Dataset Access

Hugging Face datasets can be loaded directly with the datasets library, for example `load_dataset('masakhane/masakhaner2', 'yor')`. Kaggle datasets require an account and an API token, after which they can be pulled with the kaggle command line client. Clinical datasets such as CheXpert and MIMIC-CXR require registration, credentialing and in some cases completion of a human subjects research training module, so students choosing those projects should begin the application process in week one.

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